

Supporting Information: Group-aware validation sharply reduces the apparent accuracy of extraction-category classification in public cannabis laboratory data

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This file holds the descriptive detail and the secondary tables that the main text summarises: Sections S1 to S9 and Tables S1 to S8. Section numbers without an S prefix, tables without an S prefix, Figure 1 and bracketed reference numbers refer to the main text. Every figure given here is checked against the same committed evidence logs as the main text, by `src/evaluation/audit_preprint.py`.

S1. From concentrate rows to usable rows

Most concentrate rows carry no declaration of the extraction family in `product_type` (2.1) and are dropped at that step, which is where the bulk of the reduction from concentrate rows to labelled rows happens:

Table S1. From concentrate rows to labelled rows.

concentrate or extract rows	139,714	
product_type declares the extraction family	37,310	(26.7%)
product_type does not	102,404	(73.3%)
labelled rows outside that subset	64	
total labelled rows	37,374	

The discarded majority is dominated by values that name a format rather than a method: `extracts` (34,034 rows), `concentrate`, `product inhalable` (30,586), bare `concentrate` (26,030) and `marijuana extract for inhalation` (3,901). These are not missing labels to be recovered, since the extraction family was never recorded, so imputing one would invent the dependent variable. Matching is by regular expression (`non[- ]?solvent`) rather than exact string, which also admits regional variants such as `non-solvent concentrate` (1,445 rows) and, in 64 cases, `infused pre-rolls (non-solvent)`. Those 64 rows are the reason the labelled total slightly exceeds the declared count within the concentrate subset; they are solventless by declaration but are pre-rolls rather than concentrates, they are 0.17% of the labelled data, and none names a producer, so they are outside the headline comparison.

Of the 37,374 labelled rows, 30 carry contradictory labels (2.1) and are excluded, which leaves the 37,344 usable rows of Tables S2 and S3.

Table S2. Class counts on the usable rows.

hydrocarbon	24,573	65.8%
solventless	12,771	34.2%

Grouping variables are populated as follows:

Table S3. Grouping variables on the usable rows.

producer	33,229 rows (89.0%)	96 distinct
lab	37,344 rows (100.0%)	12 distinct
strain_name	0 rows (0.0%)	0 distinct

S2. Duplicate, split and degenerate records

Rows are not samples. 1,813 usable rows (4.9%) repeat another row on product name, laboratory, producer and all 19 analyte values, but only 419 of them name a producer; the other 1,394 are producer-less rows, which have no name and match on a handful of analyte values. Separately, 1,303 samples from the producer-less laboratories are stored as two adjacent rows each, with the same laboratory and date and complementary analytes (`delta_9_thc` in one; `cbd`, `cbda`, `thca` and `total_cbd` in the other), so those laboratories' 4,115 rows describe 2,812 samples. Among the rows that name a producer, 1,071 report nothing but a `total_terpenes` of zero, in both classes, and two of the 96 producer strings are spellings of one company (260 rows).

S3. Product names that say ethanol, by year

Table S4 gives the counts behind the drift in the meaning of the label that 2.1 describes.

Table S4. Rows whose product name says ethanol, by year of test and declared category. Dated rows only: 20 of the 163 non-solvent rows carry no date.

year	solvent-based	non-solvent
2020	82	9
2021	21	1
2022	98	0
2023	13	133

S4. Analyte coverage and the total columns

The tested-versus-detected distinction of 2.2 produces two different and both meaningful coverage figures, which we separate explicitly because conflating them is easy:

Table S5. Share of the rows that name a producer with the analyte tested and with a non-zero value, by class.

analyte	tested (flag)		detected (value > 0)	
	solventless	hydro	solventless	hydro
d_limonene	85.8%	90.7%	46.1%	73.4%
beta_caryophyllene	85.8%	90.8%	53.3%	76.4%
alpha_pinene	85.8%	90.8%	38.8%	63.0%
total_terpenes	100.0%	100.0%	64.4%	83.4%

Two properties of these columns are worth stating because they bound what the features can mean.

First, `total_terpenes` is not an independent measurement. It is a near-exact sum of the ten terpenes that survive the empty-column drop ($R^2 = 0.984$ against their sum, Pearson $r = 0.992$), so it is collinear with features already in the matrix and any per-feature importance attributed to it is split credit rather than chemistry. Its residual against that sum is laboratory-specific: `CERTIFIED AG LAB` reports an exact sum (within 10^{-6}) on 99.7% of rows, `MA & ASSOCIATES` on none, which makes the residual a laboratory fingerprint in its own right. `total_thc` and `total_cbd` are derived columns of the same kind, each a fixed combination of its acid and neutral forms. Dropping the column costs 0.3 points of balanced accuracy under the all-rows random split, so nothing in this paper depends on keeping it; we keep it because the published literature does.

Second, whether `total_terpenes` is populated at all is a laboratory convention, not a per-sample decision. Every laboratory is at 0% or 100%, with no intermediate value, and the populated set is *exactly* the set of rows that carry a producer:

```
total_terpenes populated == producer present: 33,229 / 33,229 rows, no exceptions
laboratories at 100%: G3, NV CANN, DB, CERTIFIED AG, DPL NV, ERP, 374, MA
laboratories at 0%: PureVita, Cannalytics RI, Lifted Testing, Green Peaks
```

So `total_terpenes` being blank is an exact in-model indicator of "this row will be dropped by the unseen-producer scheme". That coupling is why the headline comparison in 3.2 is made on the rows that name a producer.

Finally, 101 rows carry a `total_thc` above 100%, which is impossible as a percentage; the largest is 686,400, a milligram figure in a percent column. Ninety-four come from a single laboratory. They are 0.27% of the corpus and we leave them in place rather than silently editing source measurements, but they are unit artefacts rather than chemistry and no claim here rests on them.

S5. Missingness patterns

The 4.1-point lift of the missingness-only probe (2.3) is not spread evenly. The 19 columns take only 13 distinct missingness patterns, and 7 of them are **100% one class**:

Table S6. Missingness patterns that contain a single class, rows as stored; the three largest of the seven are listed.

```
patterns that are 100% one class: 7 of 13
rows they cover: 4123 (11.04%)
n= 1495 all solventless (6 of 19 analytes reported)
n= 1303 all hydrocarbon (1 of 19 analytes reported)
n= 1303 all hydrocarbon (4 of 19 analytes reported)
```

For 11% of the usable rows the label is recoverable with certainty from which cells are blank, before any number is read. Almost all of these rows, 4,115 of 4,123, come from four laboratories (`PureVita`, `Cannalytics RI`, `Lifted Testing`, `Green Peaks`) that report a restricted analyte panel and record no producer, so the shortcut is a laboratory reporting convention and not chemistry. Much of it is an artefact of the split records of 2.1: the rows reporting one analyte and the rows reporting four are the two halves of the same 1,303 samples (1,230 of them at `PureVita`), all solvent-based, while the rows reporting six are

all non-solvent. With each pair merged the columns take 12 patterns, 6 of them pure, covering 2,820 rows (7.82%), and the in-sample lookup scores 68.8% against a 64.6% majority. The remaining 8 rows sit in three rare patterns from other laboratories and do carry a producer; 14 of the pure rows report no analyte at all and stay in the data. The 4,115 producer-less rows are also the rows the unseen-producer scheme must drop, which is why the headline comparison in 3.2 is made without them.

Laboratory and class are not degenerate, though they are far from independent. Only 373 rows, 1.0%, sit in laboratories that are more than 90% one class, and the two largest laboratories, 15,018 and 12,531 rows, both sit close to the overall 65.8/34.2 split; across all twelve the hydrocarbon share runs from 0.0% to 97.8%, and 3.2 gives the range for the eight largest.

S6. Learning rate of the alarm probes

The probes are multi-class, 11 and 33 classes, and the softmax boosting diverges on them at scikit-learn's default learning rate of 0.1: the laboratory probe's five folds score 40.6 to 69.4 and the producer probe's 10.4 to 68.7. At a learning rate of 0.05 the fold scores span 0.6 points for the laboratory and 1.6 for the producer, and those are the figures reported in 3.1. The binary extraction task is less sensitive and keeps the default: on all usable rows its random split scores 81.0% at the default rate and 79.0% at 0.05, with fold spreads of 0.4 and 0.5, and on the producer rows the held-out score is 54.0% and 54.1%.

S7. Confusion matrix and per-class recall

The pooled confusion matrix of the reported model with the producer held out (3.2) shows where its apparent skill was concentrated:

Table S7. Pooled confusion matrix of the reported model, producer held out.

unseen-producer	true hydrocarbon: 16,149 correct / 5,818 wrong
	true solventless: 3,332 correct / 7,930 wrong

Solventless recall is **32.3%** as the mean of the five per-fold recalls and $3,332 / 11,262 = \mathbf{29.6\%}$ pooled. The two differ because the folds carry very unequal numbers of solventless rows, from 951 to 4,413, and the fold with the fewest has the highest recall (48.5% against 25.3% in the fold with the most). The pooled matrix gives a balanced accuracy of 51.6%. For comparison, the per-class recalls are 89.8% and 67.1% under the random split on the same rows, and 69.9% and 39.3% on the dual-class subset grouped by producer. The model mostly answers with the majority class, and its residual skill lives on the easy side.

S8. The strain key

Section 4.3 explains why strain could not be tested cleanly. The only cultivar signal in the corpus is a key extracted from `product_name`, and Table S8 shows what it contains:

Table S8. What a strain key extracted from product names contains.

usable rows	37,344	
no cultivar key extractable at all	4,717	(12.6%)
key extracted	32,627	(87.4%)
of those, an obvious non-cultivar word (oil, thc, lab sample, acres, ethanol, raw)	7,386	(22.6% of keys)
rows yielding no usable cultivar signal	12,103	(32.4% of usable)

The 4,717 rows with no key are the 4,115 producer-less rows, which have no product name, and 602 named ones; nearly a quarter of the keys that do parse are packaging or process words rather than cultivar names.

For completeness: grouping the method task by strain key gives 66.2% balanced accuracy (+/-1.6), against 79.2% for a random split on the keyed rows, and restricting to the 214 keys that appear in both classes, 9,598 rows, gives 56.9% (+/-11.4), against 83.6%, with a macro F1 of 42.8%. An alarm probe in the sense of 2.6, predicting the key from chemistry on the 109 keys with at least 30 rows (12,829 rows), gives 32.4% against a 22.5% baseline, a lift of 9.9 points, where the majority key is the non-cultivar word `oil`. We rest no conclusion on any of the three: a 22.6%-contaminated grouping key makes the fold composition partly arbitrary. The strain question is open and cannot be answered with this corpus alone.

S9. Holding out the laboratory

The `unseen-lab` scheme of 2.4 has the highest mean of the three group-held-out schemes, 68.3%, and a fold-to-fold standard deviation of **18.6 points**: the score depends on which laboratory is held out. Across the eight held-out laboratories the hydrocarbon share ranges from 13.5% (`Cannalytics RI`, 758 rows) to 78.4% (`PureVita`, 3,137 rows), against an overall 65.8%, so holding out a laboratory shifts the test-set prior as well as the instrument, and the two effects cannot be separated with these data. With the split pairs of 2.1 merged the scheme scores 67.4% (+/-19.3). We report it for completeness and do not rest any conclusion on it.